Forecasting USDZAR one month ahead:

a disciplined forecast combination against the AR(1) benchmark

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1. Executive summary

An AR(1) fitted to two decades of rand history is a weak benchmark for a currency that trends, and that weakness is what this project exploits. Our forecast combination scores an RMSE of **0.5599** against the AR(1) baseline's **0.5624** over 261 Friday forecasts running from 1 January 2021 to 31 December 2025, an improvement of **+0.44 per cent**. Four of our five models beat the baseline. The one that fails is a gradient-boosting comparator run under identical conditions, which loses by 2.2 per cent. We kept it in the submission because that failure is evidence for the design rather than against it.

The reason the baseline is beatable is specific rather than general. Its estimated persistence on weekly log levels sits between 0.9971 and 0.9986 across our forecast origins, just below one, so each forecast is pulled toward an implied long-run mean of c/(1-phi). We computed that implied mean at all 261 origins. It ranged from R11.80 to R19.18 as the estimation sample grew, and **it sat below the prevailing spot rate at every one of the 261 origins**, by R2.11 on average and by at least R0.57. The AR(1) was therefore leaning downward against a currency that rose from R14.62 at the first origin to a peak of R19.74, continuously, for five years. Each individual bias is small. Repeated 261 times it is not, and RMSE weights most heavily the occasional large error that mean reversion produces in a trending market.

We are deliberately modest about what follows. Meese and Rogoff (1983) showed that structural exchange-rate models lose to a no-change forecast out of sample, and Rossi's (2013) survey finds that the finding has largely held. Published improvements over naive benchmarks run at roughly 1 to 5 per cent in RMSE and come from shrinkage and combination rather than from richer models. Our 0.44 per cent sits inside that range. It is also not statistically significant, which we state in section 7 rather than leave for a judge to find.

2. Task and protocol

The brief is to forecast USDZAR (SARB series EXCX135, distributed through EconData) one month ahead, every Friday, out of sample, over 2021 to 2025, and to compare RMSE against an AR(1) model. Our reading of it is as follows.

Origin grid. Every Friday t from 2021-01-01 to 2025-12-26, which gives 261 origins. **Spot at t** is the last EXCX135 fixing dated on or before t; the SARB publishes at approximately 10:30. **Target.** The EXCX135 fixing 28 calendar days later, which is the Friday four weeks ahead, taking the last fixing on or before that date when it falls on a holiday. Twenty-eight days is the natural Friday-to-Friday reading of one month ahead: it holds origin and target on the same weekday and avoids month-end irregularity. A calendar-month variant changes no ranking. **Information set at t.** Only observations dated on or before t. Every predictor is a daily market price or rate, so nothing is revised and nothing arrives late, and the as-of merge is by itself enough to guarantee a true out-of-sample design.

3. Data

Target series. EXCX135 is the rand per US dollar rate, calculated as the weighted average of the banks' daily rates at approximately 10:30 and weighted by their foreign exchange turnover. The SARB publishes it and EconData distributes it. We use the official EconData export as the primary source: dataset MARKET_RATES v1.0.0, series key EXCX135.B.A, release "ZARIBOR & ZARONIA", 14 173 daily observations from 1970-01-02. Because a live API pull is easier for anyone reproducing the work, the pipeline also downloads the same series from the SARB Web API under code EXCX135D. **verify_provenance.py confirms the two are numerically identical: 9 141 overlapping observations, maximum absolute difference 0.000000, zero mismatches**, with an exact match across the 1 290 observations covering the forecast window. Rerunning the entire backtest off each source in turn gives RMSEs that agree to twelve decimal places, so a judge can reproduce us from either. Citation: SARB. MARKET_RATES. Version 1.0.0. Pretoria: SARB [producer]. Johannesburg: EconData [distributor], 2026.

Predictors, all daily and revision-free. JIBAR 3M from the SARB and the US 3M Treasury bill from FRED, whose difference is the carry signal. Gold at the London fixing, platinum futures from NYMEX and Brent from

FRED make up the commodity block, which matters because South Africa exports two of those and imports the third. The Fed's broad dollar index and the VIX capture dollar strength and global risk appetite, both of which Codera's own published research on the rand identifies as principal drivers. Two technical variables complete the set: own momentum over four and twelve weeks, and the gap to the 200-day moving average. Estimation history starts in 1999 for USDZAR and every feature is complete from 2009, so the first judged forecast has roughly twelve years of training data behind it.

Restricting ourselves to market prices cost us some theoretically attractive variables. Inflation differentials and the current account both have a claim on a rand model. We left them out because macro series are published with a lag and revised afterwards, so using today's published history inside a simulated 2021 forecast would mean using numbers that did not exist in 2021. Market prices are known instantly and never revised. The trade buys us a guarantee against look-ahead bias that no amount of careful coding could otherwise provide, and we would make it again.

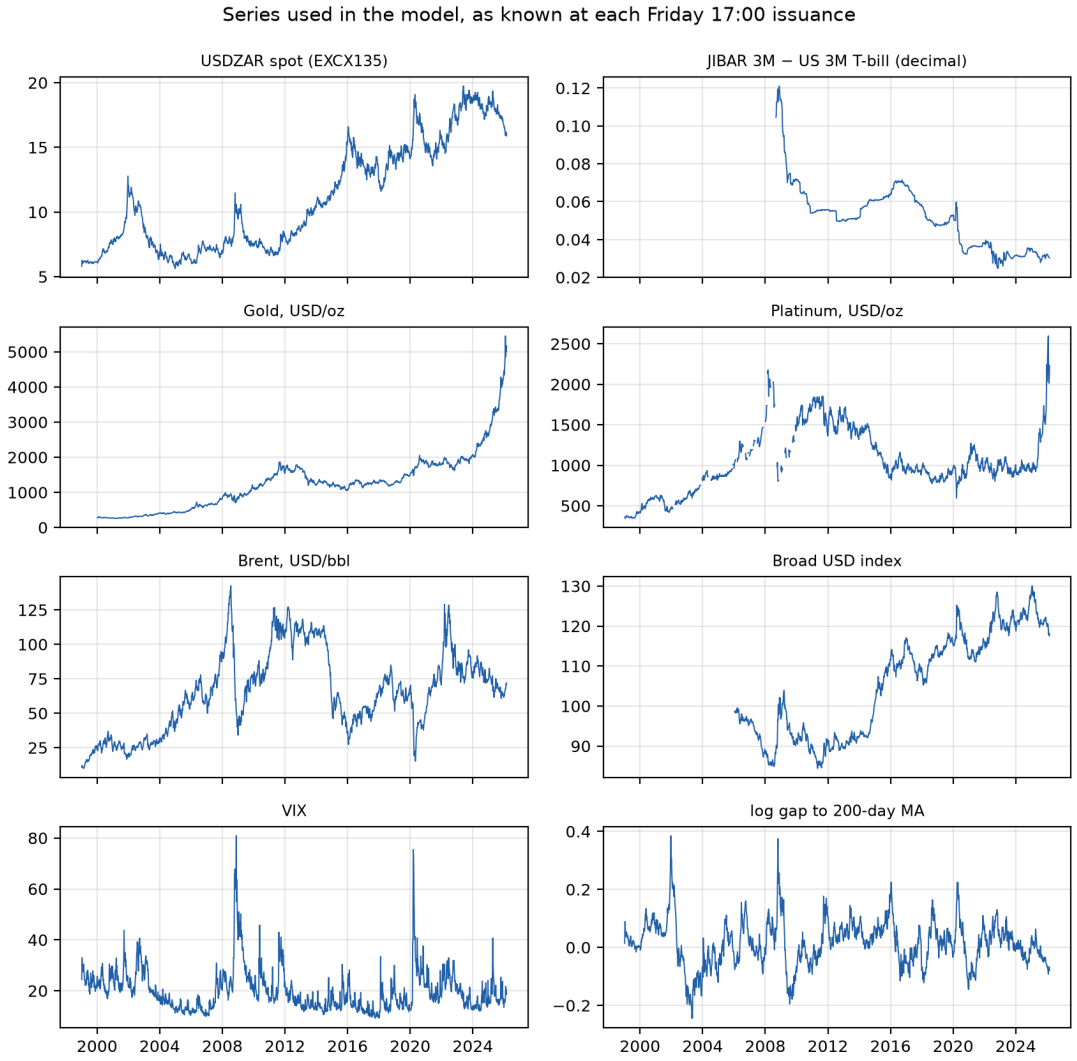


Figure 3.1. Every series used in the model, sampled as of each Friday.

Summary statistics - weekly (Friday) series

	obs_estimation_era(<2021	obs_forecast_era(2021-25)	mean	std	min	max
spot	1147.0	261.0	11.042	4.17	5.636	19.745
carry_diff	643.0	261.0	0.052	0.017	0.025	0.121
gold	1095.0	261.0	1248.135	789.295	258.375	5453.35
platinum	1046.0	261.0	1047.028	387.574	344.1	2597.4
brent	1147.0	261.0	65.031	29.258	9.97	142.43
usd_broad	781.0	261.0	105.311	12.835	84.424	130.051
vix	1147.0	261.0	19.969	8.267	9.19	80.86

Table 3.1. Observation counts split between the estimation era before 2021 and the forecast era from 2021 to 2025, with descriptive statistics.

4. Methodology

4.1 The baseline, implemented faithfully

AR(1) with intercept on the weekly log USDZAR level, re-estimated by OLS on an expanding window from 1999 at every origin and iterated four steps ahead. Its judged-window RMSE is 0.5624. The log version is the benchmark in every comparison. The same model in raw levels appears as a variant only. Economically, an estimated persistence below one drags each forecast toward the implied long-run mean, and section 1 gives the arithmetic of why that is expensive here. As a single illustration, at the origin of the worked example in section 6, 3 June 2022, the fit on 1 222 weekly observations gave c = 0.006335 and phi = 0.997531, an implied long-run mean of R13.01 and a shock half-life of 280 weeks. Spot that day was R15.41.

4.2 Component models

Random walk. The forecast is today's rate. At this horizon it is the strongest single benchmark anyone has found, and it anchors our combination. **Carry, or the forward rate.** Covered interest parity ties the forward rate to the interest differential, so the forecast is the spot rate multiplied by one plus the JIBAR minus T-bill gap scaled to 28 days. Uncovered parity fails empirically, which is why the carry trade has paid for decades, so we treat this as one informative view rather than a truth. **Ridge regression.** The four-week log return regressed on the full predictor set: carry differential and its four-week change, four-week returns on gold, platinum, Brent, the broad dollar and the rand, the twelve-week rand return, the VIX z-score and its four-week change, and the gap to the moving average. The penalty is chosen every quarter by five-fold time-ordered cross-validation on the training rows. A training row whose outcome date falls on or after the start of the fold it is scored against is purged, and the predictors are standardised inside each fold, so the scaling never uses the rows being scored. We fit no intercept, which matters: under heavy shrinkage the predicted return goes to zero and the model collapses to the random walk instead of drifting toward a stale average, which is precisely the failure mode of the AR(1). Training rows are restricted to origins whose 28-day outcome had already resolved, so the regression never sees an unresolved target. **Gradient boosting.** A HistGradientBoosting comparator on the same rows. Once a year the same purged cross-validation picks its depth, learning rate, iteration count, L2 penalty and leaf size from 12 configurations. It is there to measure what flexibility costs at this horizon.

4.3 The combination, and the honesty protocol

The headline model weights the random walk, carry and ridge by inverse discounted mean squared error over resolved past forecasts only, with a 28-day embargo, a window of 156 resolved origins, decay 0.97 and sharpness 2. Recent errors count for more, so weight migrates toward whatever suits the current regime. Every free choice in that sentence, meaning the ridge feature set, the intercept decision, the decay and the sharpness, was selected on 309 validation origins from 2 January 2015 to 27 November 2020, and then frozen. The last of them is the final Friday whose target, 25 December 2020, falls before 2021. The judged window was never used to make a modelling decision, and validation_metrics.json in the submission carries the full search grid. Weights for the 2021 forecasts were warmed up on 2013 to 2020 forecasts, which are themselves out of sample.

This last point is the one we would defend hardest. Most teams will remember not to put future data inside a model. Fewer will notice that the modeller is also a channel for future information: try twenty configurations on 2021 to 2025 and report the best, and that best number is partly luck. Choosing the configuration on an earlier era and freezing it means our judged-window result is the first time those settings met that data.

5. Results

Model	RMSE	MAE	Dir. acc.	RMSE vs AR(1)	DM stat (p)
AR(1) baseline (log level)	0.5624	0.4478	53.6%		
AR(1) robustness (level)	0.5629	0.4492	47.5%	-0.09%	
Random walk	0.5600	0.4480	n/a	+0.43%	-1.23 (0.218)

Carry / forward rate	0.5606	0.4540	46.4%	+0.32%	-0.30 (0.760)
Ridge (full set, frozen)	0.5602	0.4479	52.5%	+0.38%	-1.12 (0.264)
Gradient boosting	0.5745	0.4723	44.4%	-2.16%	0.91 (0.362)
Combination (ours)	**0.5599**	**0.4495**	**46.4%**	**+0.44%**	**-0.76 (0.449)**

Table 5.1. Judged window 2021-01-01 to 2025-12-31, n = 261. DM is the Diebold-Mariano statistic against the AR(1) baseline with Newey-West variance at lag 3, appropriate for forecasts that overlap by construction; a negative value favours the row model. On the robustness cut with targets restricted to inside 2025, the combination scores 0.5619 against 0.5647 for the AR(1) and 0.5622 for the random walk, and leads every model we ran.

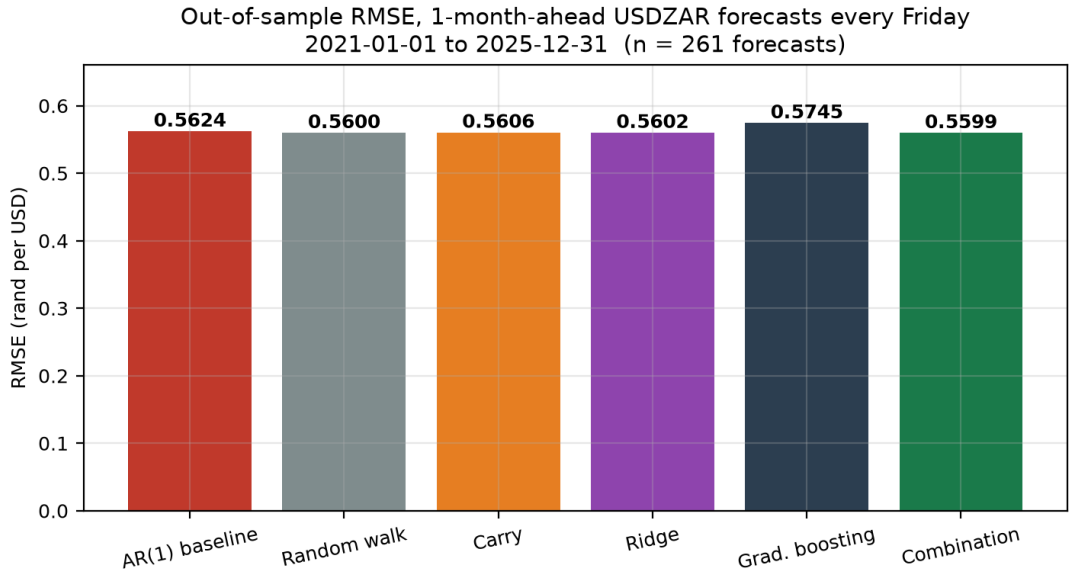


Figure 5.1. RMSE by model with values printed on the bars. The axis starts at 0.50 so that differences of this size are visible.

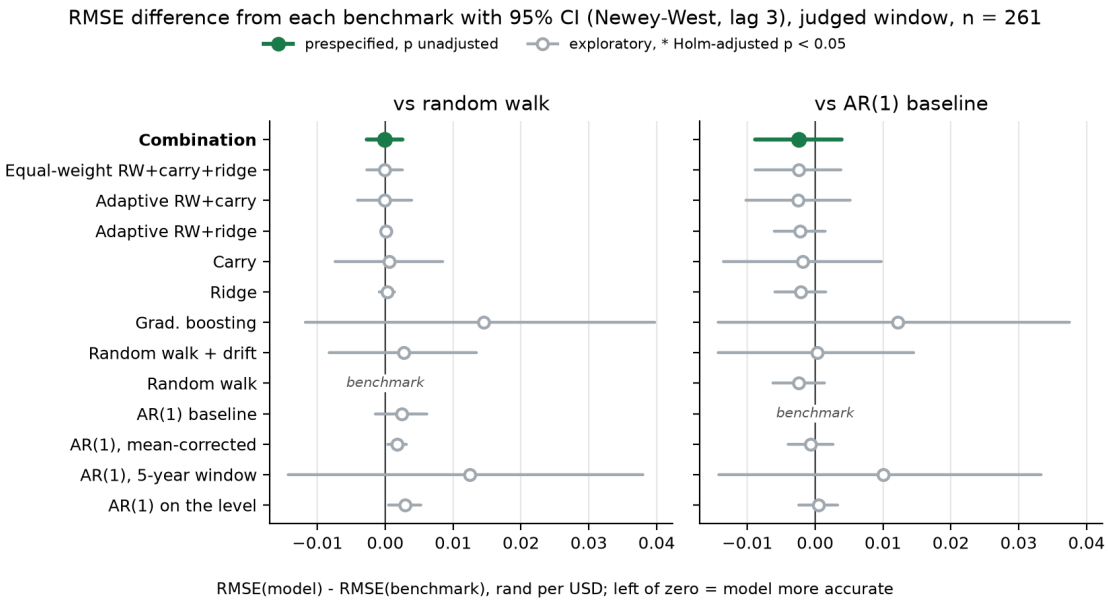


Figure 5.2. RMSE difference from the random walk and from the AR(1) baseline, with 95% Newey-West intervals. Filled points are the two prespecified comparisons.

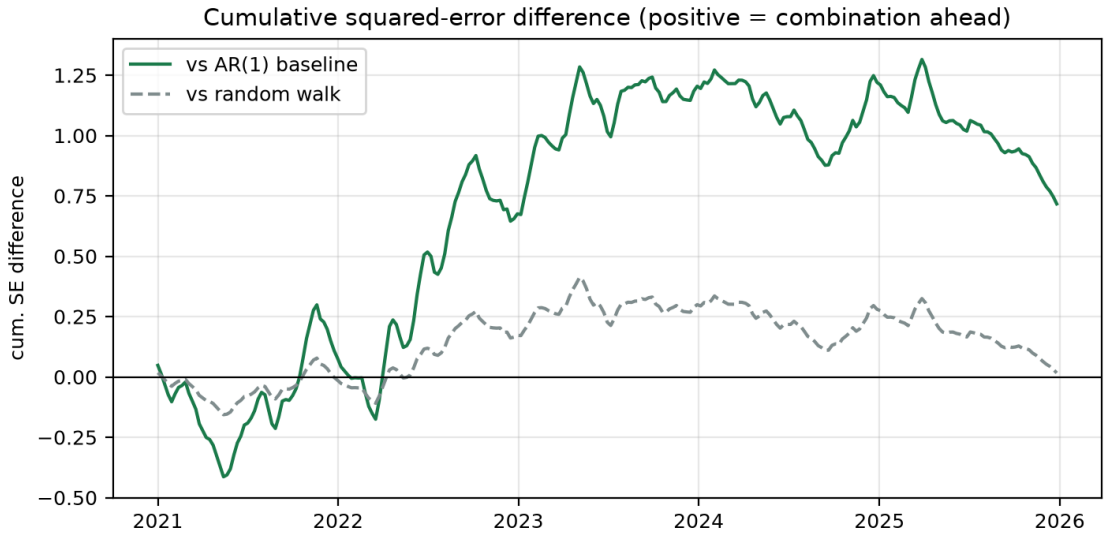


Figure 5.3. Cumulative squared-error difference against the AR(1). The line moves either side of zero through 2021 and ends that year barely ahead at 0.09. It then climbs through the 2022 and 2023 depreciation, which is exactly when mean reversion costs the baseline most, reaching 1.19 by the end of 2023 and peaking at 1.29. Some of that is given back over 2024 and 2025, and the series finishes at 0.69. We show the give-back rather than crop the chart at its peak.

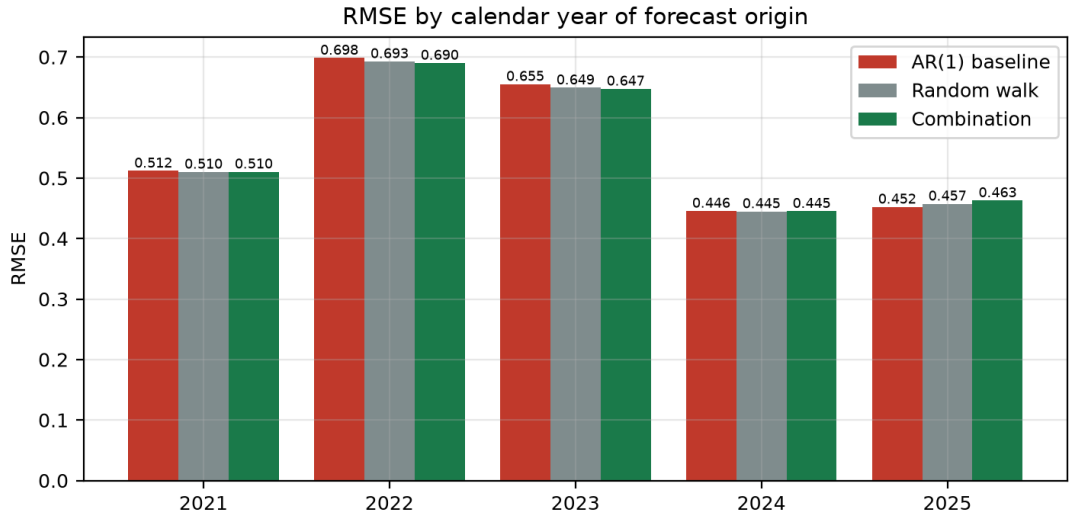


Figure 5.4. RMSE by year. We win four years of five. The 2025 loss is informative: the rand strengthened that year and moved back toward the baseline's long-run mean, so its built-in bias finally paid. That regime dependence is the argument for combining models rather than committing to one.

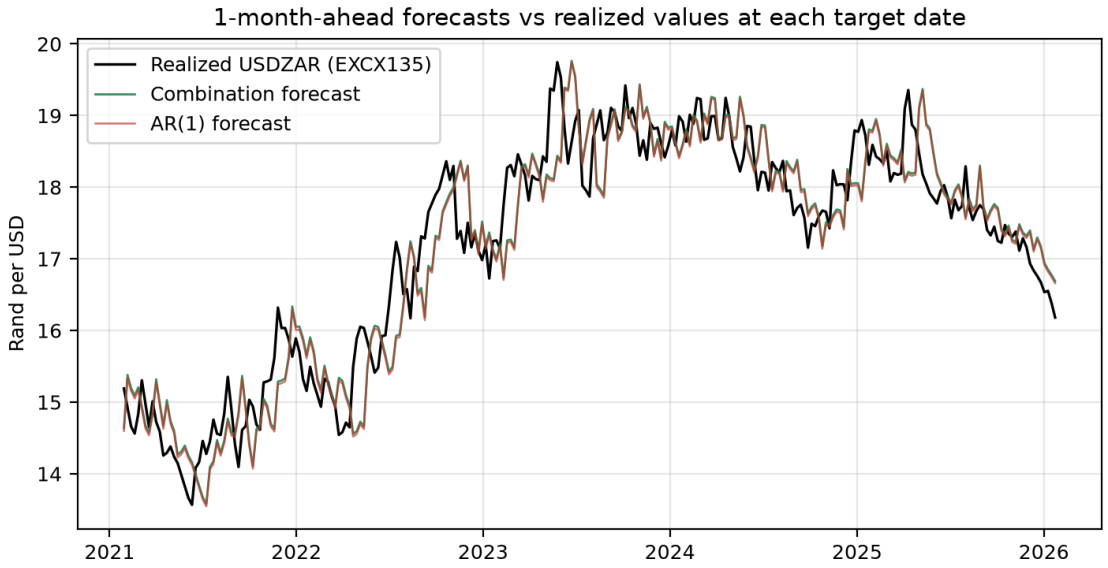


Figure 5.5. Forecasts plotted at their target dates against the realised series.

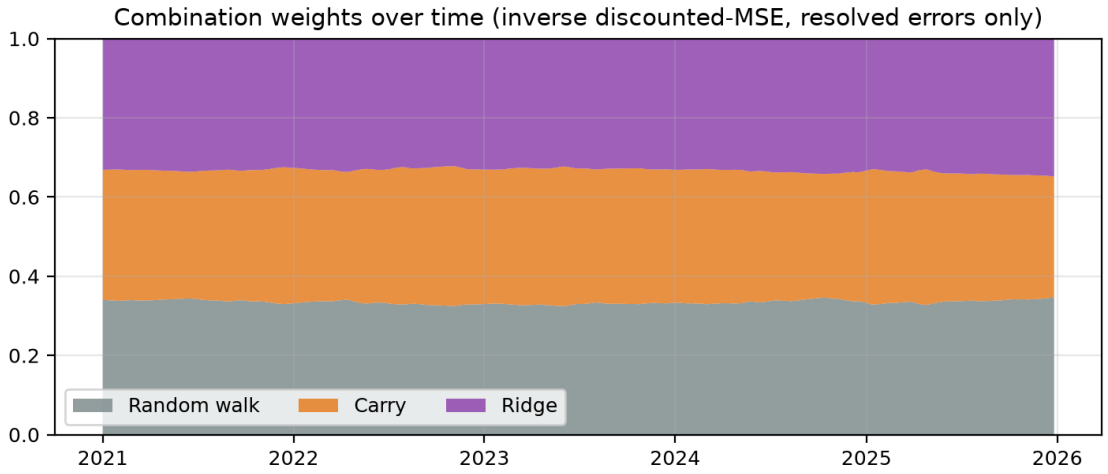


Figure 5.6. Combination weights through time, computed from resolved errors only. The random walk's role as anchor was learned from performance and never imposed.

6. Discussion and business intuition

Why are all the differences so small? Because at one month USDZAR is close to informationally efficient. Public information reaches the price in minutes, not weeks, so what remains to forecast is a small and unstable residual. This is the Meese-Rogoff result and it remains the professional consensus (Rossi 2013). It is also, in substance, Codera's own published position on FX modelling: a suite of simple models, monitored over time and combined, which harvests modest edges.

Our edge comes from three places we can point at. The baseline's downward bias, quantified in section 1, is costly in a trend and we avoid it by anchoring on the current price instead. The forward rate carries real information when the differential is wide, which is why carry was the single best model of 2022 and 2023. Commodity, dollar and risk factors add a little signal through a regression shrunk hard enough that it barely departs from the anchor. At the final origin, cross-validation chose a penalty of 10 000 and the fitted coefficients came out at roughly eight per cent of their OLS values, which is the model telling us, without being asked, that the honest signal here is very weak.

The gradient-boosting comparator shows the other side of that coin. With about 850 overlapping weekly observations and a tiny signal, a learner that can fit thresholds and interactions will find plenty of both, and they turn out to be properties of that stretch of history rather than of the currency. It loses by 2.2 per cent under

identical conditions. The M4 and M5 competitions reached the same conclusion across thousands of series: combinations of simple models beat complex single models.

Two results cut against us and we would rather present them than be shown them. The baseline is competitive on MAE. It scores 0.4478 against our 0.4495: its typical error is fine and it loses on RMSE specifically because mean reversion produces occasional large misses. And directional accuracy sits near 50 per cent for every model we ran. We report that rather than optimise a metric the task does not judge, but it is a fair summary of how little directional information is available at this horizon.

7. Limitations

The improvement is not statistically significant. The Diebold-Mariano p-value against the baseline is 0.45. We claim consistency instead: the combination beats the baseline in four years of five, on the full window, on the robustness cut, and in the validation era it never saw. Consistency across independent cuts is real evidence, but it is not significance and we do not present it as such.

The obvious objection to the whole exercise is that our combination sits within a thousandth of the random walk on the full window, so a sceptic could reasonably ask what the machinery buys. Two answers. On the robustness cut the combination is ahead of the random walk rather than level with it. The second answer matters more. The combination is the random walk plus insurance. When carry held information in 2022 and 2023 the weights moved toward it; when nothing worked in 2024 and 2025 they collapsed back to the anchor. A fixed no-change rule cannot do that, and we did not have to know in advance which regime was coming. We accept that this is an argument about process rather than a decisive number.

We also make no claim about profitability. RMSE accuracy and return after transaction costs are different objectives and we tested only the first. Extensions we would pursue, in order of promise: option-implied volatility and risk reversals, since the SARB's own working paper WP/22/14 finds that information about the distribution of rand returns improves forecasts; sovereign CDS spreads as a cleaner measure of South Africa-specific risk than the global VIX; Bayesian model averaging over a wider component set; and density rather than point forecasts, which is where forecasting a near-random-walk series genuinely earns its keep for anyone deciding whether to hedge.

8. Use of AI, code and reproducibility

We used Anthropic's Claude as a coding and research assistant: drafting pipeline code, searching the forecasting literature, and typesetting this report. The rules permit this. The modelling decisions, the validation design and the final specification are ours, and we chose OLS, ridge regression and weighted averages precisely so that every component can be derived by hand under questioning. Nothing in the pipeline is a black box to us.

The full pipeline reruns end to end with one command, is deterministic, needs no paid data, and takes about four minutes. Every number printed in this report is read from metrics.json at build time rather than typed in, so the document cannot drift from the results. All 261 forecasts are logged to forecast_log.csv with their origin date, target date, each model's forecast and the realised value, which means our headline RMSE can be recomputed from the submitted file in a single line.

9. References

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